

VLSI DESIGN FLOW | 26-07-2025

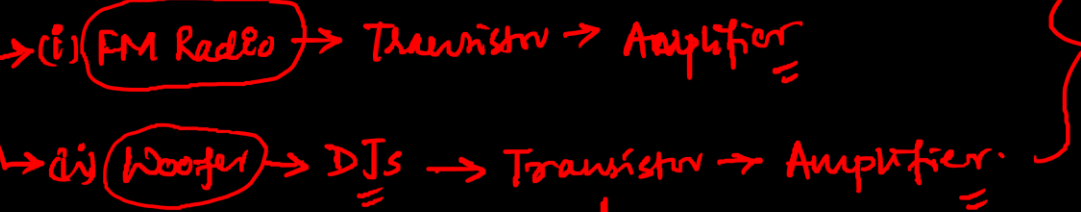
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Specialization in Microelectronics & VLSI

Application

→ SPEAKER



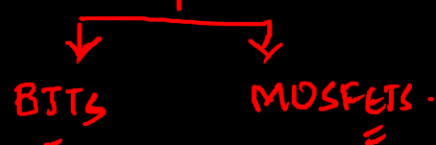
"AMPLIFIER" is a Core Analog Block.

FM Radio :-

- BW ✓ = Band of frequencies.
Gain → ∞

Speaker

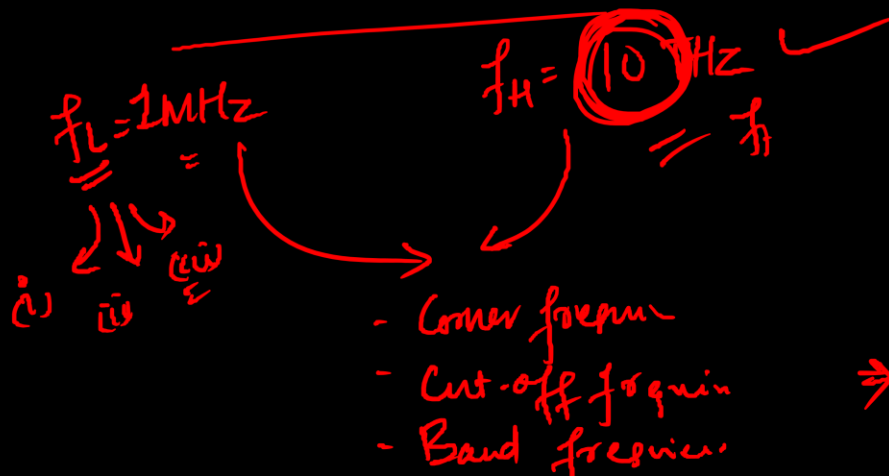
- Gain ✓
40dB
30dB



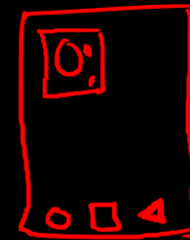
+

93.5 Hz

1 MHz



ip → signal
or
o/p → Mathematical functions



zoom

DSP → amplifier

AMPLIFIER

→ FM Radio = BW, Gain

→ Speaker = BW, Gain

GAIN
(dB)

BW
(Hz) or s

$\Rightarrow 20 \log(A_v) \rightarrow \frac{V_o}{V_{in}}$

Complex H/W
very costly

Backend design

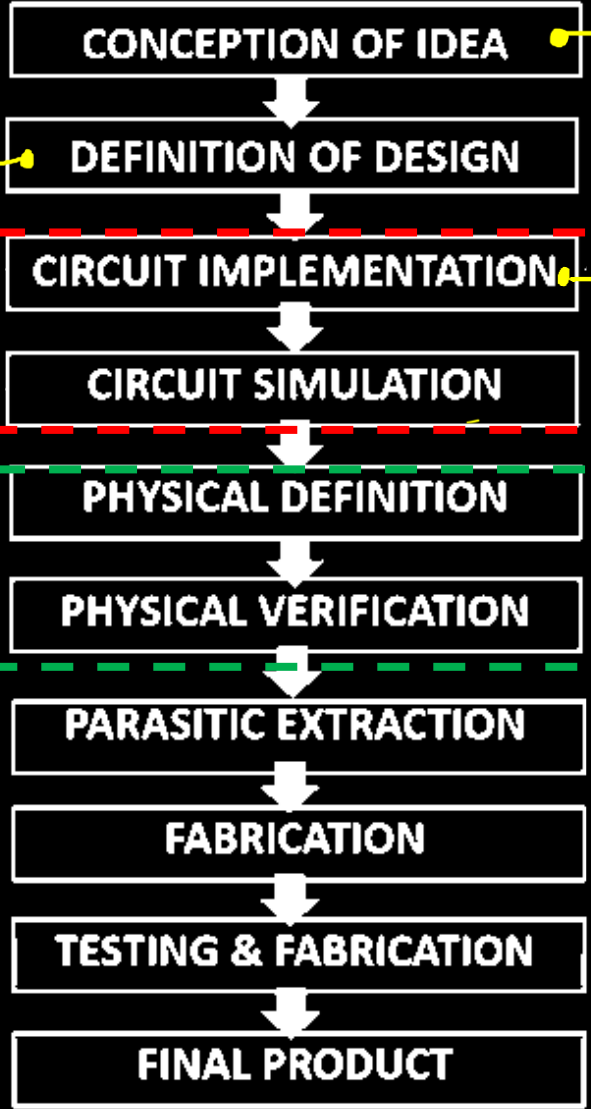
FULL CUSTOM DESIGN

Less H/W complex
FRONTEND design

SEMI-CUSTOM DESIGN

Methodology or Approach

IC DESIGN FLOW



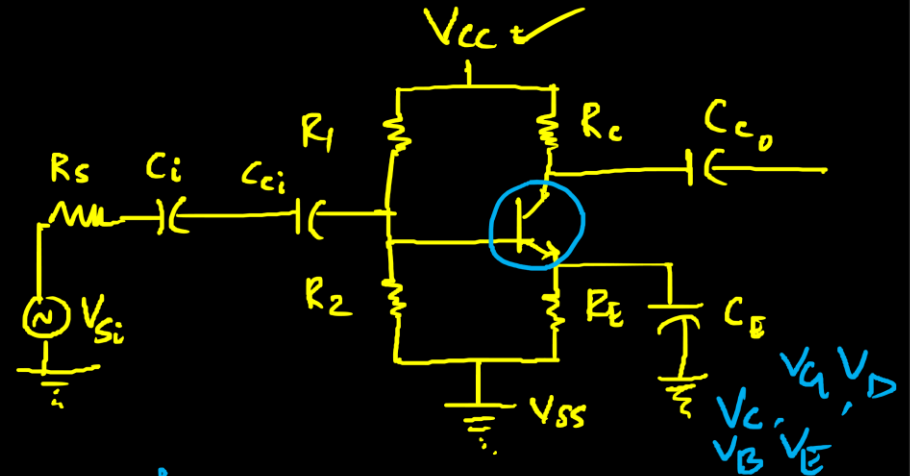
CLIENT'S segment of products =
OLED - laptop...
Smart phone
Camera
FM Radio
Speaker
Calculator

BTs ✓
MOSFETS ✓
Transistors ✓

R. → Fabrication process (Layers)
C → (Layers)

Heat sink
Power Transistor = → Fabrication process (Layers)

DEVICES to be used for the Project



- (i) BJT/MOSFET
 - gm ✓
 - Noise Margin ✓
 - Zi, Zo ✓
 - Gain ✓
 - PSD ✓
- (ii) BIASING :-
 - Voltage divider
 - Emitter follower
 - Fixed Bias
 - Stability Analysis.